

SAI Happy Farms Farm ERP — System Testing, Constraints & Acceptance Guide

Version: 1.0

Date: 04 October 2026

Status: Official Documentation

Target Audience: QA, Management, Client Sign-off Representatives

1. Testing Strategy

This guide outlines the Customer Acceptance Testing (CAT) protocol. Testing ensures the UI constraints, database synchronization, atomic transactions, and Role-Based Access Controls (RBAC) operate securely and accurately as defined by the system architecture.

The backend and frontend test suites have been verified programmatically, and this document defines the manual User Acceptance Testing (UAT) steps required for customer sign-off.

2. Test Environment

- Platform: Web / Mobile Browser
- Database: Production (or dedicated Staging) Firestore Environment
- Target Roles: Farmer, Supervisor, Admin

3. Complete Constraint Register

ID	Module	Constraint	Expected Behaviour	Status
C-01	Birds	Bird count cannot be negative	Explicit `INSUFFICIENT_BIRDS` error.	Implemented
C-02	Mortality	Mortality cannot exceed current live birds	Explicit `INSUFFICIENT_BIRDS` error.	Implemented
C-03	Culling	Culling cannot be negative	Client-side validation blocks.	Implemented
C-04	Totals	Mortality + Culling <= Live Birds	Transaction rejection.	Implemented

C-05	Feed	Feed amount must be positive	Client-side `isNaN` / negative check.	Implemented
C-06	Feed	Feed cannot exceed master stock	Explicit `INSUFFICIENT_FEED` error.	Implemented
C-07	Temperature	Must be between -10 and 60 °C	Client-side blocks form progression.	Implemented
C-08	Ammonia	Must be between 0 and 100 ppm	Client-side validation blocks.	Implemented
C-09	Duplicates	Identical daily submissions not allowed	Explicit `CORRECTION_LIMIT_REACHED`.	Implemented
C-10	Auth	User must be explicitly assigned to a farm	`FARM_NOT_ASSIGNED` thrown on read/write.	Implemented
C-11	Passwords	Passwords must be >= 8 characters	Backend validator rejects creation request.	Implemented

4. Test Results Summary

The following metrics are derived explicitly from the execution of the isolated test suites (`SECURITY\_TEST\_RESULTS.md`, `TEST\_RESULTS.md`):

- Security Testing Suites: 24 suites verified.
- Security Assertions Passed: 179 passing assertions.
- Security Failures: 0 failures.
- Manual Data Synchronization Scenario Tests: 17 tests executed successfully on live Firestore environments.
- Production Build Integrity: Frontend Vite build stripped of all diagnostic console logs successfully (Build time: 1.25s).

(Note: Specific legacy numerical claims regarding exact backend test files or XLSX import permutations are not verified from the available project evidence).

5. Customer UAT Checklist

Customers should execute the following matrices using live or staging credentials.

Authentication & Authorization Testing

Test ID	Module	Preconditions	Steps	Expected Result	Pass/Fail
---------	--------	---------------	-------	-----------------	-----------

AUTH-001	Security	Admin credentials	Log in and navigate to Users page.	Success. Page loads.	
AUTH-002	RBAC	Farmer credentials	Attempt to force-navigate to Supervisor Dashboard URL.	Access Denied. Redirected.	
SEC-001	Security	Authenticated Farmer	Execute a cross-farm read request for an unassigned farm.	HTTP 403 `AUTHORIZATION_DENIED`.	

Functional Validation Testing

Test ID	Module	Preconditions	Steps	Expected Result	Pass/Fail
FARM-001	Farm Form	Farm has 1000 birds	Attempt to submit 1001 mortality.	Blocked. `INSUFFICIENT_BIRDS`.	
FEED-001	Farm Form	Farm has 50kg feed	Attempt to submit 100kg feed used.	Blocked. `INSUFFICIENT_FEED`.	
EGG-001	Farm Form	-	Input a negative number for eggs.	Form prevents navigation to next step.	

Database Integrity Testing

Test ID	Module	Preconditions	Steps	Expected Result	Pass/Fail
SYNC-001	DB Sync	Farm has 1000 birds	Submit a valid daily report with 10 mortality.	Master count instantly reads 990.	
REV-001	Correction	Submitted V1 Report	Correct the report (V2) to 5 mortality.	Master count adjusts by delta (reads 995).	

Administrative Workflows

Test ID	Module	Preconditions	Steps	Expected Result	Pass/Fail
ADM-001	User	Admin	Create a	Validation	

		credentials	Farmer with a 7-character password.	rejects creation.	
REVERT-001	Import	Previous import batch	Leave agreement unchecked, enter Batch ID.	Revert button disabled.	
REVERT-002	Import	Previous import batch	Check agreement, enter partial ID.	Revert button disabled.	

6. Acceptance Sign-Off

By signing below, the customer representatives acknowledge that the UAT procedures have been completed, and the SAI Happy Farms Farm ERP meets the specified requirements and constraints.

Customer Representative:

Title / Organization:

Signature:

Date:

Development / Project Representative:

Signature:

Date:

Remarks:

End of Document